ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

TASK 1 REPORT

House Price Prediction Using Linear Regression

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Tool Used: Google Colab (Python)

ABSTRACT

This project builds and evaluates a Linear Regression model using the California Housing Dataset. The workflow includes loading data, exploratory data analysis, training, prediction and evaluation.

1. INTRODUCTION

Linear Regression was implemented to predict house prices using historical housing features.

2. OBJECTIVE

Understand machine learning workflow and apply Linear Regression for prediction.

3. DATASET DESCRIPTION

Features include median income, house age, average rooms, average bedrooms, population, occupancy, latitude and longitude. Target variable: Price.

4. EXPLORATORY DATA ANALYSIS

Performed dataset inspection, missing value analysis, statistics and correlation analysis.

5. MODEL BUILDING

Loaded dataset, split train/test data, trained Linear Regression model and generated predictions.

6. MODEL EVALUATION

Metrics used: MAE, RMSE and R^2 Score.

7. RESULTS

Generated prediction outputs and model evaluation results.

8. CONCLUSION

Successfully demonstrated Linear Regression using California Housing Dataset.

REFERENCES

Scikit-learn Documentation
California Housing Dataset
Google Colab